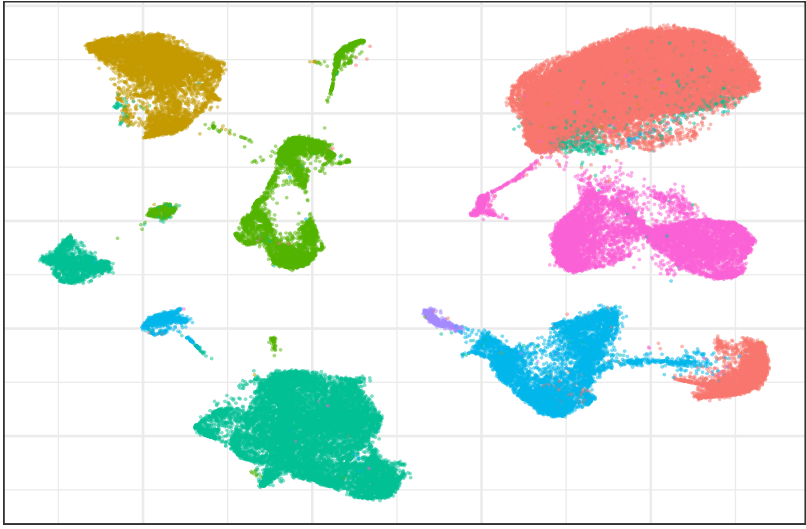


A

Human jaw (GSE303003): 65,454 cells, 4 MRONJ vs 4 cyst, 7 compartments

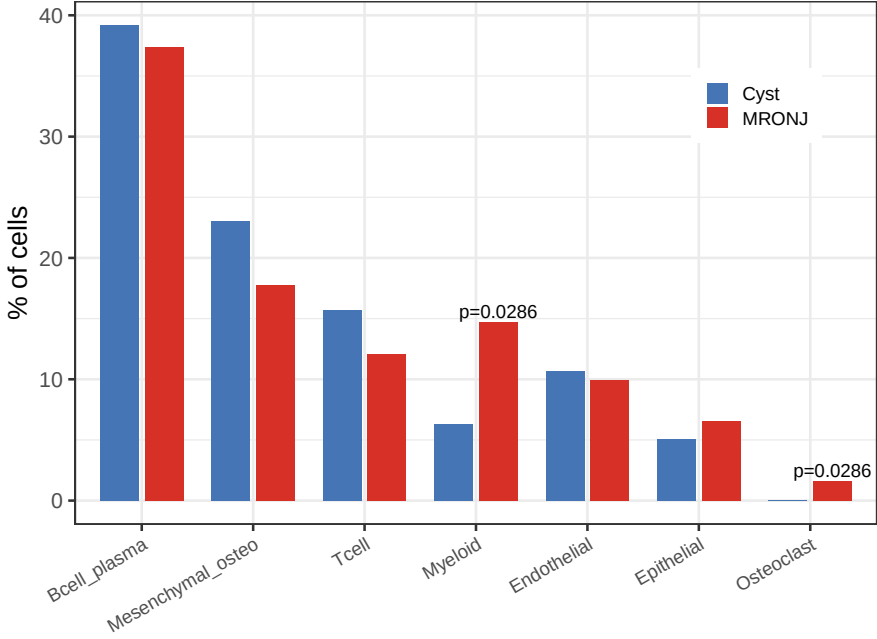


UMAP2

UMAP1

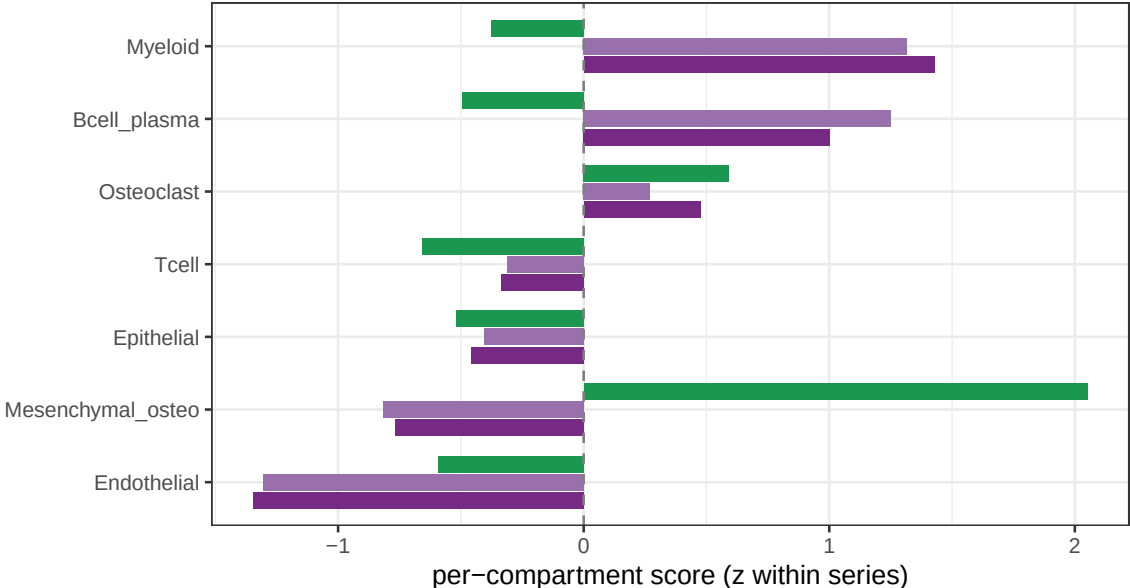
B

Compartment proportions: osteoclast & myeloid expansion in MRONJ (4)

**C**

Positive control: osteo markers peak in mesenchymal, but the L1 program does NOT

two scorings agree (Spearman rho=1); program highest in myeloid, lowest in mesenchymal/endothelial



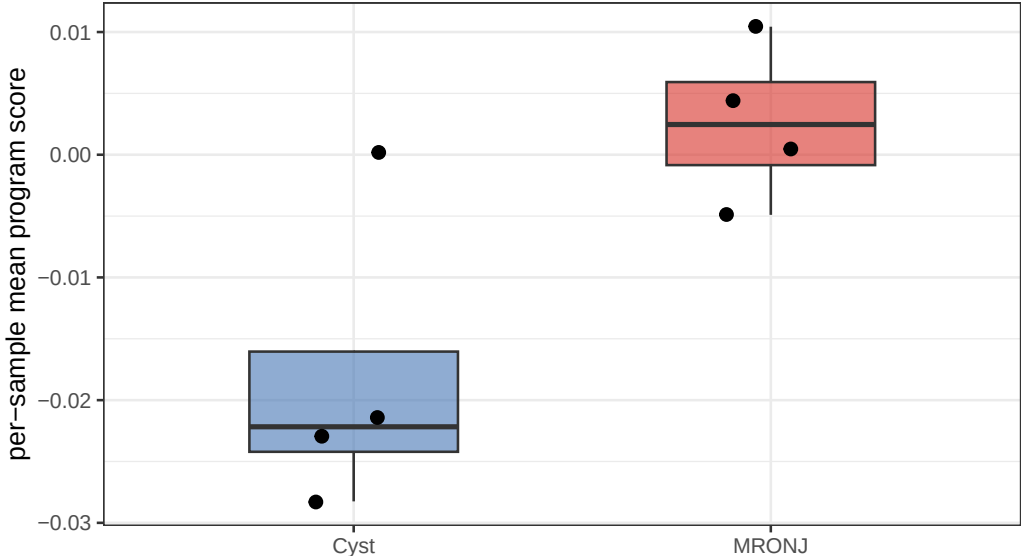
per-compartment score (z within series)

L1 program (AddModuleScore)
 L1 program (UCell, rank)
 osteo markers (RUNX2/SP7/COL1A1) — positive ctrl

D

Program NOT suppressed in MRONJ mesenchyme

whole-program sign test: 294/669 (44%) suppressed-direction, NOT > 50% (p=1); Wilcoxon p=0.057



per-sample mean program score

Cyst MRONJ